A Triadic Formalism for a Universe of Black Holes

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Abstract

We develop a complete triadic and dual-time formalization of a minimal universe consisting solely of spinless, uncharged black holes in vacuum. In this setting, black holes function as the smallest possible triadic closures: horizon-bounded domains in which energy, geometry, and information interact through a recursive triadic cycle, and in which material time emerges through mass-damped evolution. The vacuum, by contrast, possesses no closure structure, no horizon, and no material-time flow; it acts only as a transversal electromagnetic-time medium carrying the superposed Schwarzschild geometry of all closures and supporting vacuum fluctuations that drive Hawking radiation.

By integrating standard black-hole relations—including Schwarzschild radius, horizon area, holographic entropy, Hawking temperature, evaporation time, and radiative power—into the triadic architecture, we show that the minimal universe is both mathematically coherent and physically realistic. Hawking radiation arises naturally from a cross-layer interaction between t_{EM}-driven vacuum excitations, horizon geometry, and closure-internal triadic evolution governed by t_{mat}. This structure yields new derived relations linking holographic capacity, evaporation dynamics, and dual-time scaling, demonstrating that triadic unification is compatible with and enriched by established black-hole physics.

The minimal black-hole universe therefore serves as a realistic asymptotic phase of cosmic evolution and an ideal testbed for triadic unification. Its structural simplicity makes the architectural principles of the Triadic Cosmos fully explicit, while its physical grounding provides concrete, falsifiable predictions related to horizon encoding, dual-time signatures, evaporation–entropy scaling, and cross-layer Hawking dynamics. The model highlights both the power and the limitations of closure-based cosmology and opens a pathway toward extending triadic unification beyond minimal universes.

Significance Statement

This work demonstrates that a fully unified architectural description of energy, geometry, and information can be realized in a physically realistic setting. By formalizing a minimal universe containing only black holes and vacuum, we show that the structural principles of the Triadic Cosmos emerge naturally from established black-hole physics, holographic encoding, and Hawking dynamics. The model reveals that horizon-bounded closures provide a complete and self-consistent foundation for material-time evolution, while the vacuum functions as a transversal electromagnetic-time medium without requiring closure structure.

Integrating standard black-hole relations into the triadic and dual-time framework yields new structural connections between evaporation, entropy, geometry, and temporal scaling. These connections produce concrete, falsifiable predictions and demonstrate that triadic unification is not merely conceptual but empirically anchored. The minimal black-hole universe therefore serves as a realistic toy model that exposes the core mechanisms of closure-based cosmology and provides a clear pathway for extending triadic unification to richer and more complex universes.

1 Introduction

The search for a unified architectural description of physical reality has traditionally focused on dynamical frameworks: quantum field theory, general relativity, string theory, and their proposed extensions. These approaches describe the universe in terms of fields, particles, symmetries, and geometric backgrounds, but they do not provide a structural ontology capable of organizing energy, geometry, and information within a single coherent framework. The Triadic Cosmos architecture addresses this gap by introducing closures as the fundamental units of physical ontology and by formalizing the recursive interactions between energetic, geometric, and informational roles through the triadic cycle. Dual-time dynamics further distinguish transversal electromagnetic-time evolution from closure-internal material-time evolution, yielding a layered temporal structure that reflects the different ways physical systems participate in recursion.

To evaluate the coherence and empirical grounding of this architectural framework, it is useful to consider a universe in which the structural elements of the Triadic Cosmos appear in their simplest possible form. Classical cosmology predicts that the far-future universe will be dominated by isolated black holes evaporating through Hawking radiation. This late-time phase provides a natural setting for a minimal universe: one containing only spinless, uncharged black holes and vacuum. Black holes are the smallest physically realizable triadic closures, possessing horizon-bounded causal domains, holographic informational capacity, and mass-damped material-time evolution. The vacuum, by contrast, possesses no closure structure, no horizon, and no material-time flow; it acts only as a transversal electromagnetic-time medium carrying the superposed Schwarzschild geometry of all closures.

This minimal universe is not merely a conceptual simplification. It corresponds to a realistic asymptotic phase of cosmic evolution and provides a mathematically transparent environment in which the architectural principles of the Triadic Cosmos can be expressed without interference from additional fields, charges, or interactions. By integrating standard black-hole relations—including Schwarzschild radius, horizon area, holographic entropy, Hawking temperature, evaporation time, and radiative power—into the triadic and dual-time framework, we show that the minimal universe is both physically consistent and structurally complete. Hawking radiation emerges naturally from a cross-layer interaction between electromagnetic-time vacuum fluctuations, horizon geometry, and closure-internal triadic evolution.

The goal of this paper is to formalize the minimal black-hole universe within the Triadic Cosmos architecture, demonstrate its mathematical coherence, and derive testable predictions that connect triadic unification to observable astrophysical phenomena. The resulting model highlights the power of closure-based cosmology, clarifies the role of the vacuum as a non-closure entity, and provides a foundation for extending triadic unification to more complex universes containing multiple closure types and richer dynamical structure.

2 Background

Modern structural unification frameworks converge on a minimal architecture built from three irreducible roles: energy (E), geometry (G), and information (I). These roles interact through the triadic cycle $E \to G \to I \to E$, which generates emergent material time, horizon structure, and closure-dependent dynamics. Within this architecture, physical systems are modeled as *closures*: horizon-bounded, recursion-indexed domains in which the triadic cycle is well-defined.

A central insight of the Triadic Cosmos is the dual temporal architecture. Electromagnetic time (t_{EM}) is closure-transversal and universal, while material time (t_{mat}) is closure-specific and governed by entropy, recursion index, and mass-damping. Black holes provide the clearest empirical anchor for this structure: their horizon-bounded recursion, holographic information content, and Hawking-calibrated mass-damping make them canonical triadic closures. As stated in [2], "Black holes therefore serve as canonical triadic closures: systems in which horizon-bounded recursion, holographic encoding, and dual-time dynamics coincide with empirical data."

This paper applies the triadic architecture to a *minimal universe*: a vacuum containing only spinless, uncharged black holes, each characterized solely by mass and spatial position. No additional fields or charges are present. The only dynamical process is Hawking evaporation,

$$\frac{dM}{dt_{\mathrm{EM}}} \propto -\frac{1}{M^2},$$

which determines the material-time rate,

$$\frac{dt_{\mathrm{mat}}}{dt_{\mathrm{EM}}} \propto \frac{1}{M^2}.$$

In this minimal setting, Hawking radiation becomes the sole driver of triadic evolution.

Holography plays a foundational role in this construction. All physically meaningful degrees of freedom of a black hole are encoded at its horizon, consistent with the Bekenstein–Hawking entropy and extremal-surface reconstruction. The holographic encoding layer is supplied by the surviving core of string theory. As shown in [3], "What remains is a small, rigid, and mathematically coherent holographic core," consisting of extremal surfaces, entanglement wedges, conformal field theory, holographic RG flow, complexity geometry, and Calabi–Yau algebra as static encoding objects. These structures form a complete boundary-bulk encoding language without requiring extra dimensions, moduli, fluxes, or KK towers. In this reinterpretation, "string theory therefore remains deeply relevant—not as a dynamical theory of everything, but as the default holographic encoding model of physical reality."

The minimal black-hole universe is therefore not only compatible with holography—it is its simplest nontrivial instantiation. Each black hole functions as a triadic closure whose informational content is holographically encoded, whose geometry is horizon-bounded, and whose energetic evolution is governed solely by Hawking radiation. This universe also resembles the classical heat-death scenario in standard cosmology, in which all matter has decayed and only black holes remain, evaporating extremely slowly over vast timescales. The Triadic Cosmos refines this picture by noting that heat death is closure-specific rather than a global fate [1], yet the classical description remains a useful limiting case for studying horizon-dominated physics.

This paper therefore investigates a conceptually clean and structurally revealing limit of the Triadic Cosmos: a universe in which every object is a triadic closure, every temporal flow is dual-layered, every degree of freedom is holographically encoded, and every dynamical process is horizon-bounded. By applying dual time, triadic closure, holographic encoding, and black-hole thermodynamics to this minimal universe, we obtain a mathematically transparent demonstration of triadic unification.

3 Formalization of the Minimal Black-Hole Universe

This section develops the full formal structure of a universe consisting solely of spinless, uncharged black holes in vacuum. Each black hole is treated as a triadic closure: its informational content is holographically encoded at the horizon, its geometric structure is determined by Schwarzschild curvature, and its energetic evolution is governed exclusively by Hawking radiation. Dual time provides the temporal architecture, while the triadic cycle organizes the dynamical interdependence of energy, geometry, and information. The holographic encoding layer is supplied by the surviving core of string theory, which provides extremal surfaces, entanglement geometry, conformal field theory, and Calabi–Yau algebra as static encoding objects. As stated in [2], "Black holes therefore serve as canonical triadic closures: systems in which horizon-bounded recursion, holographic encoding, and dual-time dynamics coincide with empirical data."

3.1 Black Holes as Minimal Closures

In the Triadic Cosmos architecture, a physical domain qualifies as a closure only if it satisfies positivity, holographic consistency, causal accessibility, recursive stability, and full triadic-cycle closure. Black holes are the smallest physically realizable structures that satisfy all five conditions. Their horizons define finite causal domains, their informational content is holographically encoded, their geometry is positive and horizon-bounded, and their recursive evolution is governed by mass-damped material time. No smaller structure—particles, fields, or vacuum excitations—possesses a horizon or supports a triadic update cycle. Black holes are therefore the atomic units of closure in this minimal universe.

3.2 The Vacuum as a Non-Closure

The vacuum does not satisfy the architectural criteria for closure. It has no horizon, no holographic boundary, no finite causal domain, no recursion index, and no internal entropy production. It therefore possesses no triadic state (E, G, I) and no material-time flow. Instead, the vacuum functions as a transversal geometric and electromagnetic medium: it carries the superposed Schwarzschild geometry of all closures and supports the universal electromagnetic-time structure t_{EM}, but it does not participate in material-time evolution. In triadic terms,

$$E_{\mathrm{vac}} = 0, \qquad I_{\mathrm{vac}} = 0, \qquad G_{\mathrm{vac}} \neq 0,$$

with geometry induced entirely by the closures themselves. The vacuum cannot act as an observer, since observation requires closure structure, informational capacity, and a recursion index.

3.3 State Space of the Universe

The minimal universe is defined by the configuration space

$$\mathcal{C} = \{(M_i, \mathbf{x}_i) \mid i = 1, \ldots, N\},$$

where each black hole is characterized solely by its mass M_i and spatial position $\mathbf{x}_i$ in an otherwise empty vacuum. No charges, spin, fields, or additional degrees of freedom exist. The background geometry is determined entirely by the superposition of Schwarzschild metrics associated with each black hole.

3.4 Triadic State of Each Closure

Each black hole i is represented by a triadic state

$$X_i = (E_i, G_i, I_i),$$

where $E_i = M_i$ is the energetic role (mass-energy), G_i is the geometric role (Schwarzschild horizon and curvature), and $I_i = S_i = A_i/4G$ is the informational role (holographic entropy). These roles interact through the triadic cycle

$$E \to G \to I \to E,$$

which governs the recursive evolution of the closure.

3.5 Dual-Time Dynamics

The universe possesses two temporal layers:

- electromagnetic time t_{EM}, universal and closure-transversal,
- material time $t_{\mathrm{mat},i}$, closure-specific and governed by mass-damping.

The dual-time relation for each black hole is

$$\frac{dt_{\mathrm{mat},i}}{dt_{\mathrm{EM}}} = f(M_i),$$

where Hawking evaporation provides the empirical calibration:

$$\frac{dM_i}{dt_{\rm EM}} \propto -\frac{1}{M_i^2}.$$

Matching this scaling requires

$$\frac{dt_{{\rm mat},i}}{dt_{\rm EM}} \propto \frac{1}{M_i^2}.$$

Material time therefore slows dramatically as mass increases, approaching the horizon-bounded limit

$$\frac{dt_{{\rm mat},i}}{dt_{\rm EM}} \to 0^+.$$

3.6 Horizon-Bounded Causality

Each black hole defines a horizon-bounded causal domain. Internal recursion is confined to the region inside the horizon, while all observable behavior is encoded holographically at the boundary. No signals escape the interior; all accessible degrees of freedom reside on the horizon. This makes each black hole a complete closure in the triadic sense.

3.7 Holographic Encoding Layer

The informational role I_i is encoded holographically at the horizon. The encoding layer is provided by the surviving core of string theory, which remains intact under architectural and Swampland filtering. As stated in [3], "What remains is a small, rigid, and mathematically coherent holographic core." This core consists of:

- extremal surfaces (RT/HRT),
- entanglement wedges,
- conformal field theory,
- holographic RG flow,
- complexity geometry,
- Calabi–Yau algebra as static encoding objects.

These structures provide the boundary-bulk correspondence required for holographic encoding of each closure.

3.8 Triadic Update Operator

The recursive evolution of each closure is governed by the triadic update operator

$$X_{i,n+1} = R(X_{i,n}),$$

with

$$R(X_i) = (f(G_i, I_i),\ g(E_i, I_i),\ h(E_i, G_i)).$$

In the minimal universe, the only dynamical contribution to R is Hawking evaporation, which modifies E_i and thereby induces geometric and informational updates through the triadic cycle.

3.9 Vacuum Fluctuations as EM-Time Excitations

Vacuum fluctuations play a central role in the standard derivation of Hawking radiation. In quantum field theory on curved spacetime, the vacuum is not empty but exhibits transient excitations of the field modes. Near a black-hole horizon, these excitations are stretched and redshifted in such a way that the vacuum state defined by an infalling observer differs from the vacuum state defined at asymptotic infinity. The resulting Bogoliubov transformation produces a thermal spectrum of outgoing radiation. Although often heuristically described in terms of virtual particle pairs, the underlying mechanism is a geometric effect: the horizon reorganizes the vacuum mode structure.

Within the Triadic Cosmos architecture, vacuum fluctuations can be incorporated without treating the vacuum as a closure. The vacuum possesses no horizon, no holographic boundary, no recursion index, and no material-time flow. It therefore has no triadic state (E, G, I) and no internal evolution. Instead, the vacuum functions as a transversal electromagnetic-time medium. Electromagnetic time t_{EM} governs the propagation and excitation of field modes, including vacuum fluctuations. Material time t_{mat} is defined only inside closures and does not apply to the vacuum.

Vacuum fluctuations are therefore interpreted as t_{EM}-driven excitations of the geometric background. They arise because the vacuum supports electromagnetic-time evolution on a curved geometry induced by the closures. In triadic terms, the vacuum has a minimal triadic structure:

$$E_{\mathrm{vac}} = 0, \qquad I_{\mathrm{vac}} = 0, \qquad G_{\mathrm{vac}} \neq 0,$$

where G_{vac} is the superposed Schwarzschild geometry of all black holes. This minimal structure is sufficient to support vacuum excitations, but insufficient to constitute a closure.

Hawking radiation emerges from the interaction between:

- the t_{EM}-driven vacuum fluctuations,
- the horizon geometry of the closure,
- the closure's internal triadic evolution governed by t_{mat}.

The vacuum provides the excitational substrate, the horizon provides the geometric transformation, and the closure processes the resulting energy loss through its triadic update cycle. This yields a cross-layer interaction: electromagnetic-time excitations in the vacuum induce material-time evolution inside the closure.

This interpretation preserves the standard physical role of vacuum fluctuations in Hawking radiation while maintaining the architectural distinction between closures and non-closures. The vacuum does not require a full triadic cycle to exist; it requires only a minimal geometric layer capable of supporting t_{EM} excitations. Black holes remain the only triadic closures in the minimal universe, and Hawking radiation becomes the mechanism through which transversal vacuum excitations drive closure-specific triadic evolution.

3.10 Global Evolution of the Universe

The global evolution of the universe is determined by the coupled system

$$\frac{dM_i}{dt_{\mathrm{EM}}} = -\frac{k}{M_i^2}, \qquad \frac{dt_{\mathrm{mat},i}}{dt_{\mathrm{EM}}} = \frac{c}{M_i^2},$$

with constants k and c determined by horizon-scale recursion. As black holes evaporate, their material-time rate increases, accelerating their internal triadic evolution. The universe asymptotically approaches a state in which all black holes evaporate completely, leaving an empty vacuum.

3.11 Relation to Classical Heat Death

Classical cosmology describes the far-future heat-death scenario as a universe in which all matter has decayed and only black holes remain, evaporating extremely slowly. The Triadic Cosmos refines this picture by noting that heat death is closure-specific rather than a global fate [1]. Nevertheless, the minimal black-hole universe studied here corresponds precisely to the classical limiting case: a horizon-dominated universe evolving solely through Hawking radiation.

3.12 Summary

This formalization shows that a universe consisting solely of black holes provides a complete and mathematically transparent realization of triadic unification. Each black hole is a triadic closure, its informational content is holographically encoded, its geometry is horizon-bounded, its energetic evolution is governed by Hawking radiation, and its temporal structure is dual-layered. The vacuum acts only as a transversal electromagnetic and geometric medium, not as a closure. The surviving holographic core of string theory supplies the encoding layer, while the triadic cycle organizes the recursive evolution of each closure. This minimal universe therefore serves as a conceptually clean and physically meaningful demonstration of the Triadic Cosmos architecture.

4 Integration of Black-Hole Physics into the Triadic Formalism

The minimal black-hole universe can be further anchored in established black-hole physics by explicitly integrating standard relations for horizon radius, entropy, temperature, evaporation time, and radiative power into the triadic and dual-time framework. These formulas map naturally onto the triadic roles (E, G, I) and allow the construction of new composite relations that reveal the structural coherence of the minimal universe.

4.1 Schwarzschild Radius and the Geometric Role

For each spinless, uncharged black hole, the Schwarzschild radius is

$$r_s = \frac{2GM_i}{c^2}.$$

This directly defines the geometric role G_i of the closure:

$$G_i \equiv r_s(M_i),$$

and determines the causal domain and horizon-bounded region. The vacuum geometry G_{vac} is the superposition of all such Schwarzschild contributions, consistent with the vacuum's status as a non-closure that carries geometry but no triadic evolution.

4.2 Horizon Area and Holographic Entropy

The horizon area is

$$A_i = 4\pi r_s^2 = 4\pi \left(\frac{2GM_i}{c^2}\right)^2,$$

and the Bekenstein–Hawking entropy is

$$S_i = \frac{k_B c^3 A_i}{4G\hbar}.$$

In the triadic formalism, this becomes the informational role

$$I_i \equiv S_i(M_i),$$

linking $E_i = M_i$ and $G_i = r_s(M_i)$ through holographic encoding. The horizon therefore serves as the complete informational boundary of the closure.

4.3 Hawking Temperature and Internal Drive

The Hawking temperature is

$$T_{H,i} = \frac{\hbar c^3}{8\pi G M_i k_B}.$$

This provides an internal thermodynamic drive for the closure. In triadic terms, $T_{H,i}$ controls the rate at which E_i is converted into changes in G_i and I_i via Hawking radiation. Larger black holes have lower temperature, slower evaporation, and slower material-time evolution.

4.4 Evaporation Time and Dual-Time Scaling

The characteristic evaporation time for a black hole of mass M_i is

$$t_{\text{evap},i} \sim \frac{G^2 M_i^3}{\hbar c^4}.$$

In the dual-time formalism, this sets the effective lifetime of the closure in electromagnetic time t_{EM}. Combined with

$$\frac{dt_{\text{mat},i}}{dt_{\text{EM}}} \propto \frac{1}{M_i^2},$$

we obtain an effective material-time lifetime

$$t_{\text{mat,evap},i} \sim \int_0^{t_{\text{evap},i}} \frac{dt_{\text{mat},i}}{dt_{\text{EM}}}\, dt_{\text{EM}} \propto \int_0^{t_{\text{evap},i}} \frac{1}{M_i^2(t_{\text{EM}})}\, dt_{\text{EM}},$$

which links the dual-time scaling directly to the evaporation process. As the closure loses mass, its material-time rate accelerates, producing a characteristic triadic acceleration near the end of its lifetime.

4.5 Radiative Power and Triadic Update Rate

The radiative power of Hawking emission scales as

$$P_i \sim \frac{\hbar c^6}{G^2 M_i^2}.$$

This can be interpreted as the triadic update rate:

$$\frac{dE_i}{dt_{\text{EM}}} \sim -P_i,$$

with corresponding induced changes in G_i and I_i via the triadic cycle

$$E_i \to G_i \to I_i \to E_i.$$

The dependence $P_i \propto 1/M_i^2$ matches the dual-time scaling and reinforces the mass-damping interpretation: larger closures evolve more slowly in both E and t_{mat}.

4.6 Vacuum Fluctuations and EM-Time Excitations

Vacuum fluctuations, essential in the standard derivation of Hawking radiation, are incorporated as t_{EM}-driven excitations of the geometric background. The vacuum has no triadic state and no material-time flow, but supports electromagnetic-time evolution on a curved geometry induced by the closures. Hawking radiation emerges from the interaction between:

- the t_{EM}-driven vacuum fluctuations,
- the horizon geometry of the closure,
- the closure's internal triadic evolution governed by t_{mat}.

This yields a cross-layer interaction: transversal vacuum excitations drive closure-specific triadic evolution.

4.7 Derived Triadic Relations

Combining these standard formulas yields new composite relations in the triadic formalism. For example:

- A direct link between informational content and evaporation time:

  $$t_{\mathrm{evap},i} \sim \alpha\, S_i^{3/2},$$

  for suitable constant α, reflecting that larger holographic capacity implies longer closure lifetime.

- A relation between dual-time scaling and horizon area:

  $$\frac{dt_{\mathrm{mat},i}}{dt_{\mathrm{EM}}} \propto \frac{1}{M_i^2} \propto \frac{1}{r_s^2} \propto \frac{1}{A_i},$$

  showing that material time slows as holographic capacity increases.

- A triadic consistency condition:

  $$P_i \cdot t_{\mathrm{evap},i} \sim E_i,$$

  expressing that the total radiated energy over the lifetime of the closure matches its initial mass-energy.

These derived relations demonstrate that the triadic and dual-time formalism is not merely compatible with standard black-hole physics, but capable of generating new, structurally meaningful connections between horizon geometry, holographic entropy, evaporation dynamics, and temporal architecture in the minimal universe.

4.8 Phenomenological Status of the Derived Relations

The derived relations presented in this paper—such as the dual-time scaling

$$\frac{dt_{\mathrm{mat}}}{dt_{\mathrm{EM}}} \propto \frac{1}{M^2},$$

and the evaporation–entropy relation

$$t_{\mathrm{evap}} \sim \alpha S^{3/2},$$

should be understood as phenomenological structural fits rather than fundamental derivations. They arise from matching the triadic and dual-time architecture to the empirically established scaling laws of Hawking evaporation. In the minimal universe, Hawking radiation is the sole dynamical process, and its mass-damped behavior provides a natural calibration for the triadic cycle and material-time flow.

These relations therefore serve as architectural predictions: they express how the triadic roles (E, G, I) and dual-time dynamics cohere when constrained by black-hole thermodynamics. A full derivation from a deeper action principle or a more complete holographic recursion model is left for future work. The phenomenological framing adopted here is intentional: the minimal universe is a structurally transparent demonstrator of triadic unification, not a complete dynamical theory of all closure types.

4.9 Relation to Standard Holographic Black-Hole Models

The minimal black-hole universe developed in this paper is fully compatible with standard holographic descriptions of black holes, including AdS/CFT-based models, extremal-surface reconstruction, and entanglement-wedge geometry. The triadic roles (E, G, I) map directly onto familiar holographic quantities: E corresponds to mass-energy, G to horizon geometry and curvature, and I to Bekenstein–Hawking entropy and its boundary encoding.

The holographic encoding layer used in this formalism is supplied by the surviving core of string theory, consisting of extremal surfaces, conformal field theory, holographic RG flow, complexity geometry, and Calabi–Yau algebra as static encoding objects. These structures coincide with the boundary-bulk correspondence used in standard holographic black-hole models, but are interpreted here as architectural rather than dynamical components.

The main difference lies in ontology rather than physics. Standard holographic models treat black holes as geometric or thermodynamic objects embedded in a background spacetime. The triadic formalism treats black holes as closures: horizon-bounded recursion domains in which energy, geometry, and information interact through a triadic cycle and evolve under dual-time dynamics. This reinterpretation preserves all established black-hole physics while providing a structural framework that unifies holography, recursion, and temporal architecture.

In this sense, the minimal universe is a limiting case of holographic black-hole cosmology: a setting in which every object is holographically encoded, every causal domain is horizon-bounded, and every dynamical process is governed by Hawking radiation.

5 Testable Predictions of the Minimal Black-Hole Universe

Although the minimal black-hole universe is a conceptual limit of cosmic evolution, its structural simplicity makes it an ideal setting for deriving concrete, testable predictions. The triadic architecture, dual-time dynamics, and holographic encoding layer impose strict relationships between horizon geometry, evaporation behavior, informational capacity, and temporal evolution. These relationships yield empirical signatures that can be evaluated in observational astrophysics, gravitational-wave astronomy, and precision timing experiments.

5.1 Horizon-Based Encoding and Entropy Tests

The triadic formalism predicts that all physically meaningful degrees of freedom of a black hole are encoded at its horizon. Any deviation from the Bekenstein–Hawking entropy formula or extremal-surface reconstruction would violate triadic closure structure. Observable consequences include:

- horizon entropy must match $S = A/4G$ to high precision,
- black-hole shadows must reflect extremal-surface geometry,
- gravitational-wave ringdown modes must encode horizon-area changes consistently.

These predictions are testable through black-hole shadow measurements, gravitational-wave observations, and island-formula consistency checks.

5.2 Dual-Time Signatures in Low-Entropy Environments

Dual-time dynamics imply that material-time evolution slows as holographic capacity increases:

$$\frac{dt_{\mathrm{mat}}}{dt_{\mathrm{EM}}} \propto \frac{1}{A}.$$

This predicts measurable differences in physical degradation rates for systems operating in low-entropy closures versus high-entropy environments. In particular:

- deep-space probes should exhibit slower material degradation than identical systems near high-entropy astrophysical environments,
- long-baseline timing signals should reveal closure-dependent temporal divergence,
- electromagnetic coherence should remain universal across closure boundaries.

These predictions can be tested using Voyager-class probes, deep-space clocks, and pulsar timing arrays.

5.3 Evaporation–Entropy Scaling

Integrating standard black-hole relations yields a structural prediction linking evaporation time and holographic entropy:

$$t_{\mathrm{evap}} \sim \alpha S^{3/2},$$

for suitable constant α. This implies:

- larger holographic capacity implies systematically longer closure lifetime,
- evaporation dynamics must follow a predictable entropy-driven trajectory,
- deviations from this scaling would falsify the triadic evaporation model.

This prediction can be evaluated through semiclassical modeling of black-hole populations and consistency checks in numerical relativity.

5.4 Cross-Layer Hawking Dynamics

Hawking radiation arises from the interaction between:

- t_{EM}-driven vacuum fluctuations,
- horizon geometry,
- closure-internal triadic evolution governed by t_{mat}.

This cross-layer mechanism predicts:

- the Page curve must follow holographic entanglement predictions,
- information must be preserved through horizon encoding,
- bulk information loss would violate triadic closure structure.

These predictions are testable through island-formula calculations, entanglement-wedge reconstruction, and gravitational-wave observations of late-time black-hole dynamics.

5.5 Consistency of Radiative Power and Triadic Update Rate

The radiative power of Hawking emission,

$$P \sim \frac{1}{M^2},$$

matches the dual-time scaling and triadic update rate. This implies:

- evaporation accelerates in material time as mass decreases,
- late-time black-hole dynamics must exhibit triadic acceleration,
- the total radiated energy over the lifetime of the closure must match its initial mass-energy.

These predictions can be tested through gravitational-wave ringdown modeling and semiclassical evaporation simulations.

5.6 Summary

The minimal black-hole universe yields a set of concrete, falsifiable predictions grounded in horizon physics, holographic encoding, dual-time dynamics, and triadic closure structure. These predictions connect the architectural principles of the Triadic Cosmos to observable astrophysical phenomena and provide a clear empirical pathway for evaluating the coherence and validity of the unification framework.

6 Open Questions and Limitations

Although the minimal black-hole universe provides a structurally transparent and physically grounded testbed for triadic unification, several open questions and limitations remain. These arise primarily from the treatment of the vacuum, the minimality of the model, the role of closures in cosmic evolution, and the interpretation of heat death within the Triadic Cosmos architecture.

6.1 Vacuum Structure and Non-Closure Status

The vacuum plays a central role in the minimal universe: it carries the superposed Schwarzschild geometry of all closures and supports electromagnetic-time evolution, including vacuum fluctuations that drive Hawking radiation. However, the vacuum is not a closure and possesses no triadic state, no horizon, no recursion index, and no material-time flow. This raises several open questions:

- To what extent can vacuum fluctuations be fully described as t_{EM}-driven excitations without invoking additional quantum-field structure?
- Does the vacuum require a minimal triadic substrate to exist, and if so, how should this be formalized without violating the closure criteria?
- Can the vacuum support more complex transversal dynamics in non-minimal universes, and how would these interact with closure-based evolution?

These questions highlight the need for a deeper architectural understanding of the vacuum as a non-closure entity that nevertheless participates in horizon-level physics.

6.2 Minimality of the Universe Model

The universe studied in this paper is intentionally minimal: it contains only spinless, uncharged black holes and vacuum. This makes the model structurally clean and mathematically tractable, but also limits its scope. In particular:

- the absence of matter, radiation, and fields eliminates many dynamical processes present in realistic cosmological settings,
- the lack of interactions between closures (other than gravitational) restricts the range of possible triadic evolutions,
- the model does not address closure formation, merger dynamics, or early-universe processes.

These limitations are inherent to the minimal design but suggest that future work should explore non-minimal universes containing multiple closure types and richer dynamical structure.

6.3 Heat Death and Closure-Based End States

Classical cosmology predicts that the universe will eventually reach a heat-death state dominated by isolated black holes evaporating through Hawking radiation. The minimal universe corresponds to this limiting case, but the Triadic Cosmos architecture reframes heat death as a closure-specific rather than global phenomenon. This raises several open questions:

- Is heat death the unique asymptotic end state of closure-based universes, or can alternative closure configurations persist indefinitely?
- Can new closures form in late-time cosmology through quantum or holographic processes?
- Does the complete evaporation of all closures represent a true termination of cosmic evolution, or does the vacuum itself admit further structural transitions?

These questions suggest that heat death may not be the only or final endpoint of triadic evolution.

6.4 Architectural Boundaries of Triadic Unification

The minimal universe demonstrates that triadic unification is compatible with black-hole physics, holographic encoding, and dual-time dynamics. However, several architectural boundaries remain:

- The triadic cycle is fully realized only inside closures; its extension to non-closure entities remains unclear.
- Dual-time dynamics rely on Hawking evaporation; alternative drivers of material-time evolution in non-minimal universes require further study.
- The holographic encoding layer is static in the minimal universe; its dynamical behavior in universes with multiple closure types is not yet formalized.

These limitations indicate that the minimal universe is a foundational demonstration rather than a complete description of triadic unification.

6.5 Summary

The minimal black-hole universe provides a conceptually clean and physically realistic setting for triadic unification, but its simplicity also reveals several open questions regarding vacuum structure, closure dynamics, cosmic endpoints, and architectural generalization. Addressing these questions will be essential for extending the Triadic Cosmos beyond minimal universes and toward a full unified description of physical reality.

7 Conclusion

This paper has developed a complete triadic and dual-time formalization of a minimal universe consisting solely of spinless, uncharged black holes in vacuum. In this setting, black holes serve as the smallest possible triadic closures: horizon-bounded domains in which energy, geometry, and information interact through the triadic cycle, and in which material time emerges through mass-damped recursion. The vacuum, by contrast, possesses no horizon, no recursion index, and no material-time flow. It functions only as a transversal electromagnetic-time medium carrying the superposed Schwarzschild geometry of all closures.

The resulting universe is both conceptually minimal and physically realistic. Classical cosmology predicts that the far-future universe will be dominated by isolated black holes evaporating through Hawking radiation. The model developed here corresponds precisely to this limiting case, but expressed through the structural architecture of the Triadic Cosmos. Hawking radiation emerges naturally from the interaction between t_{EM}-driven vacuum fluctuations, horizon geometry, and closure-internal triadic evolution. This cross-layer mechanism provides a concrete realization of dual-time dynamics and demonstrates how transversal vacuum excitations drive closure-specific material-time evolution.

Integrating standard black-hole relations—including Schwarzschild radius, horizon area, holographic entropy, Hawking temperature, evaporation time, and radiative power—anchors the triadic formalism in established physics. These formulas map cleanly onto the triadic roles (E, G, I) and yield new derived relations linking holographic capacity, evaporation dynamics, and dual-time scaling. The consistency of these relations shows that the triadic and dual-time architecture is not merely compatible with black-hole physics, but capable of revealing new structural connections within it.

This minimal universe therefore provides a powerful and transparent demonstration of triadic unification. It shows that the surviving holographic core of string theory can function as a complete encoding layer, that dual-time dynamics can be grounded in Hawking evaporation, and that closures can replace global spacetime as the fundamental units of physical ontology. By reducing the universe to its simplest horizon-bounded components, the model makes the architectural principles of the Triadic Cosmos fully explicit and empirically anchored.

In this sense, the minimal black-hole universe is not only a conceptual toy model but a realistic asymptotic phase of cosmic evolution. Its structural clarity and physical grounding make it an ideal testbed for triadic unification, demonstrating how energy, geometry, and information cohere within a single architectural framework and how horizon-bounded physics can serve as the foundation for a unified description of reality.

A Vacuum Closure Possibility in Non-Minimal Universes

In the minimal black-hole universe studied in this paper, the vacuum is not treated as a closure. It has no horizon, no holographic boundary, no recursion index, and no material-time flow. Its triadic state is therefore minimal:

$$E_{\text{vac}} = 0, \qquad I_{\text{vac}} = 0, \qquad G_{\text{vac}} \neq 0,$$

with geometry induced entirely by the closures. This interpretation is appropriate for a universe in which the vacuum contains no fields, charges, or dynamical degrees of freedom beyond electromagnetic time and curvature.

However, the architectural criteria of the Triadic Cosmos do not forbid the possibility of vacuum closures in more complex universes. In settings where the vacuum possesses emergent horizons, non-trivial informational structure, or recursion-supporting degrees of freedom, it may satisfy the closure conditions of positivity, holographic consistency, causal accessibility, and recursive stability. Examples include universes with dynamical fields, cosmological horizons, or holographic vacuum domains.

The minimal universe therefore represents a specific instantiation of the triadic architecture rather than a universal constraint. Vacuum non-closure is a structural choice appropriate for the simplest possible universe, but alternative cosmologies may admit vacuum closures with full triadic states and material-time evolution. Exploring such possibilities is an important direction for future work.

B Comparison to Alternative Quantum Gravity Frameworks

The minimal black-hole universe formalized in this paper is structurally grounded in the triadic architecture, dual-time dynamics, and holographic encoding. Although the model is developed within the Triadic Cosmos framework, it is instructive to compare its structure to alternative approaches in quantum gravity. This appendix outlines how a universe containing only spinless, uncharged black holes would be represented in several major frameworks and assesses the degree of compatibility with the triadic formalism.

Loop Quantum Gravity (LQG)

Loop Quantum Gravity describes spacetime as a network of quantized geometric excitations. In LQG, black holes are modeled through isolated horizon frameworks and quantum geometry operators acting on spin networks. A minimal universe containing only black holes would be represented as a collection of isolated horizons embedded in a largely trivial spin network.

Such a universe is compatible with the triadic formalism in several respects:

- horizon area quantization aligns with the informational role $I_i = A_i/4G$,
- isolated horizons provide a natural boundary for recursion and causal structure,
- the absence of bulk fields matches the triadic interpretation of the vacuum as a non-closure.

However, LQG lacks a holographic encoding layer and does not provide a dual-time architecture. Material-time evolution would need to be introduced phenomenologically, and the triadic cycle would not arise naturally from spin-network dynamics. Compatibility is therefore partial but structurally promising.

Causal Set Theory

Causal Set Theory models spacetime as a discrete partially ordered set. Black holes appear as regions with characteristic causal structure deficits or horizon-like boundaries. A minimal universe of black holes would be represented as a set of disjoint causal domains with no additional field content.

This picture aligns well with the triadic interpretation of closures as horizon-bounded causal domains. The vacuum, which in the triadic formalism has no recursion index and no triadic state

$$E_{\text{vac}} = 0, \qquad I_{\text{vac}} = 0, \qquad G_{\text{vac}} \neq 0,$$

corresponds naturally to regions of the causal set with trivial order structure.

Causal Set Theory does not provide holographic encoding or dual-time dynamics, but its causal architecture is highly compatible with the closure ontology. A triadic reinterpretation of causal sets appears feasible.

Asymptotic Safety

Asymptotic Safety treats gravity as a renormalizable quantum field theory governed by a nontrivial UV fixed point. Black holes are modeled through semiclassical solutions corrected by running couplings. A minimal universe of black holes would be described by scale-dependent Schwarzschild geometries with no additional matter fields.

This framework is compatible with the triadic formalism at the level of geometry and evaporation dynamics. The running of Newton's constant modifies horizon area and evaporation time, but the triadic roles (E, G, I) remain well-defined. Dual-time dynamics would need to be introduced phenomenologically, as Asymptotic Safety does not provide a layered temporal structure.

Emergent Spacetime and Quantum Information Approaches

Approaches in which spacetime emerges from quantum entanglement—such as tensor networks, quantum error correction models, and complexity geometry—are closely aligned with the holographic encoding layer used in this paper. A minimal universe of black holes would be represented as a set of boundary-encoded regions with trivial bulk entanglement.

This is highly compatible with the triadic architecture:

- holographic entropy matches the informational role I_i,
- extremal surfaces and entanglement wedges coincide with the encoding layer,
- the vacuum as a non-closure corresponds to regions with no entanglement structure.

Dual-time dynamics could be interpreted as a separation between transversal entanglement evolution and closure-internal recursion. This makes emergent-spacetime approaches the closest match to the triadic formalism.

Summary

Across alternative quantum gravity frameworks, the minimal black-hole universe is broadly compatible with geometric, causal, and holographic structures. The main differences concern temporal architecture and recursion: dual-time dynamics and the triadic cycle are unique to the Triadic Cosmos. Nevertheless, the core features of the minimal universe—horizon-bounded domains, holographic entropy, and vacuum non-closure—appear in multiple frameworks, suggesting that the triadic interpretation is structurally robust and widely interoperable.

C Compatibility with Existing Theory-of-Everything Frameworks

If the Triadic Cosmos provides a correct minimal unification architecture, then any physically viable Theory of Everything (TOE) must be compatible with its structural principles. This follows from the fact that the triadic roles (E, G, I), horizon-bounded closures, holographic encoding, and dual-time dynamics represent the minimal set of empirically grounded features that appear across all established approaches to quantum gravity and high-energy physics.

A physically meaningful TOE must reproduce:

- horizon-bounded causal domains,
- holographic entropy and boundary encoding,
- geometric–informational duality,
- mass-damped or entropy-driven temporal evolution,
- recursion or update dynamics within finite causal regions,
- a vacuum sector that may or may not support closure structure.

The minimal black-hole universe developed in this paper contains all of these features in their simplest possible form. As a result, any TOE that is grounded in empirical physics must be able to represent a universe consisting solely of spinless, uncharged black holes in vacuum. This requirement is not restrictive; it is a consistency condition. A TOE that cannot model such a universe would fail to reproduce well-established black-hole thermodynamics, holographic entropy, and horizon geometry.

The compatibility analysis in Appendix B demonstrates that major quantum-gravity frameworks—including Loop Quantum Gravity, Causal Set Theory, Asymptotic Safety, and emergent-spacetime approaches—can represent a minimal black-hole universe, even if they differ in ontology or dynamical structure. This suggests that the triadic architecture captures a set of structural invariants that persist across diverse theoretical models.

In this sense, the Triadic Cosmos does not compete with existing TOE proposals; it provides the minimal architectural substrate that any physically correct TOE must satisfy. The minimal black-hole universe therefore serves as a universal benchmark: a limiting case that tests whether a candidate theory can reproduce the essential geometric, informational, and causal features of horizon-dominated physics.

D Relation to Observational Cosmology

Although the minimal black-hole universe represents an asymptotic and highly idealized cosmological limit, several aspects of its structure connect directly to observational cosmology. This appendix outlines how the triadic formalism interfaces with empirical data from black-hole astrophysics, gravitational-wave astronomy, late-time cosmology, and horizon-scale observations.

Black-Hole Populations and Mass Distributions

Current observations reveal a diverse population of astrophysical black holes, ranging from stellar-mass objects to supermassive black holes in galactic centers. While the minimal universe contains only spinless, uncharged black holes, its structural relations—particularly those linking mass, horizon area, entropy, and evaporation time—apply to all observed black holes.

The triadic roles (E, G, I) map directly onto measurable quantities:

- E corresponds to mass-energy inferred from orbital dynamics and gravitational-wave signals,
- G corresponds to horizon radius and curvature measurable through black-hole shadows,
- I corresponds to holographic entropy proportional to horizon area.

These correspondences allow the triadic formalism to interpret observed black-hole populations as non-minimal realizations of closure structure.

Gravitational-Wave Signatures

Gravitational-wave observations from LIGO, Virgo, and KAGRA provide direct access to horizon-scale dynamics. Ringdown modes encode changes in horizon area, and merger remnants exhibit mass-damped evolution consistent with the triadic update cycle.

The minimal universe predicts:

- horizon-area changes must follow triadic consistency conditions,
- ringdown frequencies must reflect geometric updates in G_i,
- late-time behavior of merger remnants must exhibit mass-damped temporal evolution.

These predictions can be tested using current and future gravitational-wave catalogs.

Black-Hole Shadows and Horizon Geometry

Observations from the Event Horizon Telescope (EHT) provide direct constraints on horizon geometry. The triadic formalism predicts that all physically accessible degrees of freedom reside on the horizon, consistent with holographic encoding.

In particular:

- shadow size and shape must reflect extremal-surface geometry,
- deviations from Schwarzschild or Kerr predictions would signal non-minimal closure structure,
- horizon-scale observables must match the informational role $I_i = A_i/4G$.

These observations provide a direct empirical anchor for the geometric role G_i.

Late-Time Cosmology and Heat-Death Scenarios

The minimal black-hole universe corresponds to the classical heat-death limit of cosmology, in which all matter has decayed and only black holes remain. Observational cosmology provides several lines of evidence supporting this trajectory:

- the accelerating expansion of the universe,
- the decreasing matter density over cosmic time,
- the long-term dominance of black-hole evaporation in late-time dynamics.

The triadic refinement—that heat death is closure-specific rather than global—remains compatible with these observations. The minimal universe therefore represents a physically meaningful extrapolation of current cosmological trends.

Page Curve and Information Preservation

Recent advances in holographic entanglement and island-formula calculations provide strong evidence for information preservation in black-hole evaporation. The triadic formalism predicts:

- the Page curve must follow holographic entanglement predictions,
- information must be encoded at the horizon throughout evaporation,
- bulk information loss is incompatible with closure structure.

These predictions align with current theoretical and semi-observational results in quantum gravity.

Summary

Although the minimal black-hole universe is an asymptotic and idealized model, its structural features connect directly to observational cosmology. Black-hole shadows, gravitational-wave signals, horizon geometry, and late-time cosmological trends all provide empirical anchors for the triadic roles (E, G, I), dual-time dynamics, and holographic encoding. The minimal universe therefore serves not only as a theoretical demonstrator of triadic unification but also as a physically grounded model with clear observational relevance.

E Mathematical Tightening of the Derived Relations

This appendix provides compact derivations and consistency checks for the structural relations used in the main text. These derivations are not intended as fundamental proofs but as dimensional and phenomenological confirmations that the triadic scaling laws are compatible with standard black-hole thermodynamics.

Dual-Time Scaling from Hawking Evaporation

The Hawking evaporation rate for a spinless, uncharged black hole is

$$\frac{dM}{dt_{\mathrm{EM}}} \propto -\frac{1}{M^2}.$$

Material time t_{mat} is defined as mass-damped evolution inside the closure. If the triadic update rate is proportional to the Hawking-driven change in $E = M$, then

$$\frac{dt_{\mathrm{mat}}}{dt_{\mathrm{EM}}} \propto \left| \frac{dM}{dt_{\mathrm{EM}}} \right| \propto \frac{1}{M^2}.$$

Thus the dual-time relation follows directly from the evaporation scaling:

$$\frac{dt_{\mathrm{mat}}}{dt_{\mathrm{EM}}} \propto \frac{1}{M^2}.$$

Evaporation Time as a Function of Holographic Entropy

The standard evaporation time is

$$t_{\mathrm{evap}} \sim \frac{G^2 M^3}{\hbar c^4}.$$

The holographic entropy is

$$S = \frac{k_B c^3}{4G\hbar} A = \frac{k_B c^3}{4G\hbar} 4\pi r_s^2 \propto M^2.$$

Thus $S \propto M^2$ implies $M \propto S^{1/2}$, and therefore

$$t_{\mathrm{evap}} \propto M^3 \propto S^{3/2}.$$

This yields the structural relation used in the main text:

$$t_{\mathrm{evap}} \sim \alpha S^{3/2}.$$

Relation Between Dual-Time Scaling and Horizon Area

Since

$$r_s = \frac{2GM}{c^2}, \qquad A = 4\pi r_s^2 \propto M^2,$$

we have

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \propto \frac{1}{M^2} \propto \frac{1}{A}.$$

Thus material time slows as holographic capacity increases.

Triadic Consistency Condition

The radiative power of Hawking emission scales as

$$P \sim \frac{\hbar c^6}{G^2 M^2}.$$

Multiplying by the evaporation time,

$$P \cdot t_{\text{evap}} \propto \frac{1}{M^2} \cdot M^3 \propto M,$$

which matches the energetic role $E = M$ of the closure.
This confirms the triadic consistency condition:

$$P \cdot t_{\text{evap}} \sim E.$$

Summary

These compact derivations show that the structural relations used in the triadic formalism follow directly from standard black-hole thermodynamics. A full dynamical derivation from a triadic action or holographic recursion model is left for future work.

F Triadic Cosmos Ecosystem Context

This paper is part of the *Triadic Cosmos* ecosystem, which develops a unified conceptual and computational framework spanning physical ontology and modular architectures for intelligent systems. All materials, extended notes, and evolving documentation are available in the public repository.

`https://github.com/triadic-cosmos`

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